

3

Johnson, Project Apollo, and the Politics of Space Program Planning

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Lyndon Johnson was a difficult, imperious character with a penchant for overheated rhetoric and big political plans. He left a record of landmark social gains and disastrous public failures, always using his presidential office to the hilt. Civil rights, voting rights, and Medicare alone are enough to give him a place in twentieth-century American history with Franklin Roosevelt, the greatest domestic reform president in the national experience. Johnson's spectacular failure in Vietnam would suffice to label him as one of the worst foreign policy leaders in the country's history. In the nearly thirty years since he left the White House we have not come to terms with this political giant. Indeed, this generation of Americans probably never will. Memories of Johnson's many transgressions against the national self-esteem remain too fresh to allow a sufficiently detached assessment of the man's impact on the country's life. One hopes this will change in time. For we need to see Johnson's career not as a chance to indulge our sense of moral superiority but as an opportunity to gain an understanding of many subjects crucial to the nation's past and future.

Space policy seems as good a place as any to begin. The major part Johnson played in shaping the country's space program in the 1950s and 1960s did not provoke then, nor does it now, the kind of controversy we associate with his war on poverty, the Great Society, and the Vietnam War. Moreover, Johnson's views of space tell us a great deal about his whole political career: that is, about his priorities and the means he used to achieve them. More specifically, four considerations determined Johnson's thinking about space policy in 1957-69: national security, personal political and party gain, domestic social advance, and budgetary constraints. None of these concerns, however, operated to the exclusion of the others. To be sure,

at one time or the other each of these goals became the dominant motive in determining LBJ's response to changing circumstances at home and abroad, but the other aims were never far from his mind. Yet however much Johnson's motives altered over time in dealing with space matters and however much his levels of support for space exploration rose and fell, especially in the last years of his presidency, he deserves to be remembered as the elected official who did as much, if not more, for space exploration than any other American political leader in this century.

Johnson and Early Space Policy

Sputnik I and *II*, the Soviet Earth satellites launched in October and November 1957, spurred Johnson's initial interest in fostering an aggressive American space program. His first aim was to advance the country's missile technology and eliminate a "missile gap" between the United States and the Soviet Union. Second, he believed that promoting a space program was good politics for himself and his party. "The issue [*Sputnik*] is one which, if properly handled, would blast the Republicans out of the water, unify the Democratic party, and elect you President," George Reedy, a principal Senate aide, told him. "I think you should plan to plunge heavily into this one." Johnson saw the political advantage to himself and the Democratic party in seizing the space issue. But he feared a witch-hunt that might undermine confidence in the country's military strength and encourage the belief that we could not meet the Soviet challenge.

Johnson genuinely put the national security issue first in trying to design a response to the Soviet's demonstrated superiority in the space race. During the winter of 1957-58, as chairman of an Armed Services subcommittee on preparedness, he held hearings on how the United States could produce better missiles at a faster rate. The hearings' sole objective, he declared, was securing the defense of the United States; he had no interest in finger-pointing or assessing blame for past mistakes and wished to use the past strictly as a guide for future action. John Steele, *Time's* congressional correspondent, told his editors that Johnson would "run a good investigation" that would serve a useful purpose. There would be no "political witch-hunt. Johnson knows that a good investigation is the only kind that will satisfy anyone, and in the end bring credit to anyone. . . . Here, as downtown [at the White House], there is a sense of urgency, of consideration of the national interest."

Yet Johnson was not simply a selfless patriot. As one official at the Department of Defense (DOD) said, "No sooner had *Sputnik's* first beep-beep

been heard—via the press—than the nation's legislators leaped forward like heavy drinkers hearing a cork pop." The facts emerging from Johnson's investigation demonstrated the Eisenhower administration's ineptness in mounting an effective missile and space program. It also allowed Johnson to identify himself as the country's leading congressional advocate of a stepped-up effort in space. He dominated the hearings, introducing witnesses, leading cross-examinations, and making himself the principal spokesman to the press. In January 1958, he told the Senate Democratic caucus that "control of space means control of the world" and urged his party colleagues to sign on to a greatly expanded space effort. Later that month, at the conclusion of the hearings, he persuaded his subcommittee to issue seventeen recommendations that, without being overtly partisan, showed Johnson and the Democrats as pushing the Eisenhower administration into what they thought was essential for the national well-being.

The journalists Rowland Evans and Robert Novak described Johnson's handling of the *Sputnik* crisis as "a minor masterpiece." Without involving himself in a direct collision with Eisenhower, Johnson used the space issue to damage the White House and benefit himself and the Democrats. Yet at the same time, he served the nation by propelling it into the space age. Specifically, he took the leading role in Congress in sponsoring legislation to create a National Aeronautics and Space Administration (NASA). Although Johnson's aides did much of the work on the space bill, he played a significant part in shaping NASA's organization. While understanding that the military would have a large say in any space program, *he argued successfully* for making NASA a civilian agency. Such a move would avoid service rivalries and satisfy political demands for peaceful uses of space. "The space program was a paramilitary operation in the cold war, no matter who ran it," the historian Walter A. McDougall wrote, but civilian control headed off a significant imbalance between the services and met the political needs of American officials at home and abroad.¹

The Vice President as Space Czar

Johnson's election to the vice presidency in 1960 gave him a continuing role in space policy. This defied the tradition of consigning a vice president to the outer fringes of power. The office of vice president, Thomas R. Marshall, Woodrow Wilson's vice president observed, "is like a man in a cataleptic state. He cannot speak. He cannot move. He suffers no pain. And yet—he is conscious of all that goes on around him." "The chief embarrassment in discussing his [the vice president's] office," Wilson wrote,

"is that in explaining how little there is to be said about it, one has evidently said all there is to say." Johnson, who had a lifelong aversion to being anything but top dog, later described the vice presidency as "nothing," saying "I detested every minute of it." Daniel Patrick Moynihan remembered looking into Vice President Johnson's eyes and thinking, "This is a bull castrated very late in life."²

Though Johnson's vice presidency was largely a ceremonial job, his role in space matters went beyond what a vice president normally did. In 1961, President Kennedy persuaded Congress to amend the 1958 space law to make the vice president, instead of the president, the chairman of the Space Council, an advisory group that President Eisenhower *had largely ignored* in 1958–61. Kennedy had no intention of letting Johnson eclipse him on a matter given high public visibility by Soviet space shots, but he was eager to use Johnson's expertise on something of vital national concern. Moreover, in giving Johnson some prominence as an architect of America's space program, Kennedy was making him a political lightning rod. Should an effort to catch and pass the Soviets in space technology fail or suffer a well-publicized defeat, Johnson would be out front taking some, if not much, of the heat.³

Yet Johnson eagerly accepted the risk. He saw American achievements in space as vital to the cold war contest with the Soviet Union. The Soviets' more advanced space program in 1957–61 persuaded Kennedy, Johnson, and millions of Americans that they were falling behind not only in missile technology but also in the global competition for "hearts and minds."

Consequently, in April 1961, after a Soviet cosmonaut became the first man to orbit Earth and the failure at the Bay of Pigs had embarrassed the United States, Kennedy asked Johnson to make "an overall survey of where we stand in space. Do we have a chance of beating the Soviets by putting a laboratory in space, or by a trip around the moon, or by a rocket to land on the moon, or by a rocket to go to the moon and back with a man? Is there any other space program which promises dramatic results in which we could win?" Johnson replied that the Soviets were ahead of us "in world prestige attained through technological accomplishments in space." And other nations, identifying space gains as a reflection of world leadership, were being drawn to the Soviets. A strong effort was needed at once to catch and surpass the Soviets if the United States were to win "control over . . . men's minds through space accomplishments." Johnson recommended "manned exploration of the moon" as "an achievement with great propaganda value." "The real 'competition' in outer space," he said, was

between the communist and free enterprise social systems. The control of outer space was going to "determine which system of society and government [would] dominate the future. . . . In the eyes of the world, first in space means first, period; second in space is second in everything." When people complained about the cost of space exploration, Johnson replied: "Now, would you rather have us be a second-rate nation or should we spend a little money?"⁴

Kennedy needed no prodding from Johnson to make the case for some dramatic space venture. At the end of May 1961, he told a joint session of Congress:

If we are to win the battle that is now going on around the world between freedom and tyranny, the dramatic achievements in space which occurred in recent weeks [a suborbital flight by astronaut Alan Shepard] should have made clear to us all . . . the impact of this adventure on the minds of men everywhere, who are trying to make a determination on which road they should take. . . . Now it is . . . time for this nation to take a clearly leading role in space achievement, which in many ways may hold the key to our future on earth.

Kennedy asked the country to commit itself to the goal of landing an American on the Moon and returning him safely to Earth before the decade was out.⁵

Yet Kennedy worried that a highly publicized American space effort that ended in failure would further damage the nation's prestige and inflict a political wound that could jeopardize his hold on the presidency. Shepard's flight had encouraged Kennedy's hopes that America might catch and pass the Soviets, but he remained concerned about future mishaps. In June, when Shepard drove with President Kennedy, Johnson, and Newton Minow, head of the Federal Communications Commission, to speak before the National Convention of Broadcasters, Kennedy poked Johnson and said: "You know, Lyndon, nobody knows that the Vice President is the Chairman of the Space Council. But if that flight had been a flop, I guarantee you that everybody would have known that you were the Chairman." Everyone laughed, except Lyndon, who looked glum and angry, especially after Minow chimed in, "Mr. President, if the flight would have been a flop, the Vice President would have been the next astronaut."⁶

The possibility that he would be a political sacrificial lamb for a faulty space effort did not dampen Johnson's enthusiasm for a manned mission to the Moon. His commitment rested partly on his faith in liberal nationalism, the ability of government to ensure economic and social progress through the use of its largesse. For Johnson, whose whole career had been built on the assumption that federal monies, well spent on infrastructure,

social programs, and defense, could serve the national well-being, but especially in the less affluent South, the space program was a splendid way to serve the country's defense, expand the domestic economy, and advance scientific understanding. In 1963, when criticism from academics, journalists, and political conservatives began to be heard against "the moonoggle," Johnson told Kennedy, "The space program is expensive, but it can be justified as a solid investment which will give ample returns in security, prestige, knowledge, and material benefits." During a plane trip as vice president to visit various space installations around the United States, Johnson gave an hour-long "very impassioned talk" to Minow on the virtues of communications satellites in both advancing education in underdeveloped countries and educational television in the United States.⁷

Johnson also saw other, more selfish benefits flowing from the space program. Convinced he was backing a winner, he made strong efforts to identify himself with every aspect of its work. Not only did he crisscross the country in publicized visits to space installations, he also gave a series of "factual space reports to the public" on the work of NASA and his Space Council. The ostensible objective was to educate the country, but it had the added advantage of keeping his name in the news.⁸

Then there were the pork-barrel gains that served the economic interests of Texas and the South and strengthened his political hold on the state and the region, especially at a time when his support of civil rights for African Americans was undermining it. Although Johnson denied any part in the selection of southwestern companies receiving Apollo or Moon program contracts or in shifting half of the space operations from Cape Canaveral in Florida to a command center in Houston, Senator George Smathers (D-FL) knew better. "He and I had a big argument about it, big fight," Smathers remembered. "Johnson tried to act like he didn't know. . . . It never has made sense to have a big operation at Cape Canaveral and another big operation in Texas. But that's what we got, and we got that because Kennedy allowed Johnson to become the theoretical head of the space program." Indeed, with Robert Kerr of Oklahoma, a Johnson friend, running the Senate Space Committee; Texas Congressmen Overton Brooks and Olin Teague the House counterparts; Albert Thomas, another Texas representative, chairing the Appropriations Committee; and James Webb, Johnson's nominee, directing NASA, the southwest generally and Texas in particular profited most from Kennedy's accelerated space program.⁹

In 1962, when lobbyists and congressmen from outside the South began to complain about a southwest monopoly on NASA contracts, Kennedy made Richard L. Callaghan, a congressional staffer, an assistant

administrator to James Webb. Callaghan's job was to arrange for a more equitable distribution of contracts, which would relieve congressional pressure on Kenny O'Donnell, the president's liaison to Congress, and find out whether Kerr and Johnson were pulling strings at NASA for their friends. As Callaghan later told Robert Sherrod, a *Time-Life* reporter: "Kenny O'Donnell wasn't only interested in getting the contractors off his back. He wanted to satisfy himself about the Kerr-Johnson influence on the Space Agency. He wanted to find out who was getting what—wanted to satisfy himself that the organization was honest." VERY INTERESTING," Sherrod wrote in a note to himself. "OBVIOUSLY JFK PUT O'DONNELL UP TO PLANTING CALLAGHAN, BUT HOW TO PROVE THAT THE PRESIDENT WAS SUSPICIOUS OF THE BIGGEST TEXAS WHEELER DEALER OF THEM ALL, AND OF THE 'KING OF THE SENATE,' WHO SUCCEEDED HIM AS CHAIRMAN OF THE SPACE COMMITTEE?" According to what Sherrod later learned from O'Donnell, there was no evidence to prove any wrongdoing by anyone at NASA. Nor could they find anything on Johnson that might have made him a potential liability to the Kennedy administration. As Johnson himself later ruefully noted, "The damn press always accused me of things I didn't do. They never once found out about the things I did do."¹⁰

Johnson's thousand days as vice president justifiably enhanced his reputation as someone who saw substantial national benefits flowing from expanded U.S. space efforts. It also demonstrated his effectiveness in building a national consensus for a space program. As Webb later told a BBC interviewer,

When President Kennedy asked him [LBJ] to prepare a memorandum as to what our space programme should be, . . . he called in some businessmen. . . . Then he called in Wernher von Braun and General Schriever from the Air Force and a large number of technical people and sort of had hearings. As we approached the end of that, he called in the political leaders . . . in Congress and he in effect said to them: "We ought to go forward but we don't want to go forward unless you are going to commit yourself to stay with us." . . . So he developed this commitment of certain leaders . . . and this you see made it a lot easier for the rest of the country to come along. They saw that these very powerful, responsible people, both political people in the Congress and business people from outside, believed this should be done, then we will accept it and go forward.¹¹

A Space Advocate in the White House

During his first year as president, from November 1963 to November 1964, Johnson pushed hard to keep the space effort on track. Although deter-

mined to keep his first budget under \$100 billion in order to win passage of Kennedy's \$11 billion tax cut pending before Congress, Johnson agreed to increase NASA spending by \$150 million to \$5.25 billion. "Our plan to place a man on the moon in this decade remains unchanged," he told Congress in January 1964. "It is an ambitious and important goal. In addition to providing great scientific benefits, it will demonstrate that our capability in space is second to no other nation's." But, he emphasized, "we cannot reach this goal without sufficient funds. There is no second-class ticket to space."¹²

At the same time, Johnson's decision to press ahead with Project Apollo—the U.S. Moon landing—rested less now than in 1961–63 on considerations of national security. In May 1963, he had declared, "I do not think this generation of Americans is willing to go to bed each night by the light of a Communist moon." During the first year of his presidency, he remained eager to beat the Soviets in the space race, but a U.S. missile buildup under Kennedy, JFK's success in the Cuban missile crisis in 1962, and the Nuclear Test Ban Treaty in 1963 had eased concerns about a missile gap and fears that we had fallen behind the Soviets in military might and scientific research. On the day of his assassination, Kennedy himself had intended to say "that there was no longer any fear that a Communist lead in space would become the basis of military superiority."¹³

Some worries about these matters remained, but during the first half of 1964 Johnson put greater emphasis on working out cooperative agreements with the Soviet Union to explore outer space. "President Johnson has apparently lost his enthusiasm for the Soviet-American space race," the *New York Herald Tribune* reported in June 1964. Earlier in the year, the president had sent the deputy administrator of NASA, Hugh L. Dryden, to Geneva "to seek agreements for a 'widening area' of cooperation in space with Moscow." Judging from National Security action memoranda in 1964, Johnson was clearly eager for less competition and more cooperation with the Soviets in space. As the astronaut and later Senator John Glenn saw it, the Congress was no longer so easily moved to increase space spending by appeals to the Soviet threat. "The anti-Russian theme had worn out," Glenn remembered. Johnson, ever sensitive to congressional moods, saw the need to press the case for space exploration on other grounds.¹⁴

A more compelling consideration with Johnson, especially at the start of his presidency, was to carry out Kennedy's agenda. Johnson had to confront the grief and despair many people felt over the assassination of a beloved leader and their antagonism toward someone who, however much

he might identify himself with JFK, seemed like a usurper, an unelected, untested replacement for the man the country now more than ever saw as more suitable for the job. In the first days of his presidency, only 5 percent of the public felt they knew very much about Johnson, while 67 percent said they knew next to nothing about him. Seventy percent had doubts about how the country would "carry on without" Kennedy. Seeing an essential need for continuity, for reassurance that the new president would be faithful to the previous administration's ends and means, Johnson made fulfillment of Kennedy's promise to put a man on the Moon and safely return him to Earth by 1970 one of his major priorities.¹⁵

Apollo: A Great Society Initiative

Johnson also believed that the Apollo mission made excellent economic and political sense. Landing a man on the Moon would not only reaffirm America's superiority over the Soviet Union and honor Kennedy's memory, it would also spur both immediate and long-term economic growth and gain the administration considerable political credit with the public. Less than a month after becoming president, Johnson was pressing NASA to use its resources to help Wisconsin and Minnesota expand "their research and engineering capabilities." Webb, who was a savvy politician in his own right and understood perfectly the importance of tying NASA to specific economic benefits around the country, laid plans to double NASA's "activity" in both states. More importantly, he kept close track of how NASA affected the nation's economy and took every opportunity to apprise Johnson of these gains. In a 1965 report to the president, for example, he pointed out that in the previous year 94 percent of NASA's "procurement dollars" had gone to 20,000 private U.S. industrial companies: \$331 million had been spent in 120 cities in 22 states with high unemployment rates, and as many as 750,000 people worked directly or indirectly on NASA-related business.¹⁶

Johnson understood that much more than pork-barrel spending would result from NASA's efforts generally and Project Apollo in particular. To be sure, as a seasoned politician with a keen appreciation for federal largesse, he greatly valued the economic and political gains coming to localities and his White House from NASA's spending in Florida, Alabama, Texas, Oklahoma, California, and other states around the country. But he also placed considerable value on the long-term national advances that NASA's work seemed likely to produce. As Webb told him, NASA's accomplishments were leading to the development of "new materials . . . [and] new structures" as

well as "complex electronic, mechanical and chemical systems. . . . This new technology . . . is bringing with it revolutionary change in the way of making and testing things, not only for space systems, but for innumerable other non-space services, processes and materials."

Because these benefits were essentially abstractions, Webb took pains to enumerate the many more concrete returns flowing from NASA's research and development. He told Johnson NASA had something to offer to law enforcement in terms of data processing and communication systems; to the construction industry through NASA developed materials; to pollution control through the development of an outlook whereby the Earth's air and water are beginning to be viewed as finite resources operating as closed systems; to transportation of people in and out of the inner city through research on short-haul aircraft; to improvement of economic opportunities for all citizens by stimulating business through new inventions and transfers of space technology to industry; and to a richer life by development of techniques making possible cheaper, lighter, and more reliable television sets and other electronic items for use in the home.¹⁷

For Johnson, the work of space exploration was part of a "Great Society," a larger vision he enunciated in May 1964. In a speech at the University of Michigan, he appealed to the best in the American temperament:

For a century we labored to settle and to subdue a continent. For half a century we called upon unbounded invention and untiring industry to create an order of plenty for all of our people. The challenge of the next half century is whether we have the wisdom to use that wealth to enrich and elevate our national life, and to advance the quality of our American civilization. . . . For in your time we have the opportunity to move not only toward the rich society and the powerful society, but upward to the Great Society. . . . It is a place where men are more concerned with the quality of their goals than the quantity of their goods.

To reach this promised land, Americans would have to pledge themselves to a crusade for excellence. "For better or for worse, your generation has been appointed by history . . . to lead America toward a new age," he insisted. "Will you join in the battle to build the Great Society, to prove that our material progress is only the foundation on which we will build a richer life of mind and spirit?"¹⁸

"An obvious component of this [Great Society] theme," one White House aide told Edward Welsh of the Space Council, "is the vast array of implications of our present Research and Development activity." Webb understood perfectly what Johnson had in mind. "I know of no area," he told the president, "where the inspirational thrust toward doing everything

required of a great society can be better provided on a proven base of competence, and with so many practical additional benefits to be derived, than through the space program. . . . The space program lies in your first area of building the great society, for it is truly an imaginative new program based on new ideas and new capabilities." Early in 1965, after becoming vice president, Hubert Humphrey echoed Webb's point in a speech at the Goddard Memorial Dinner:

Let me assure you that the Great Society envisioned by President Johnson is not one limited to the fight against poverty, ignorance, disease, and intolerance. The Great Society requires, in addition, an urgent quest for excellence, for intellectual attainment, for crossing new frontiers in science and technology. Let me emphasize that an adequately funded, well-directed space program is an integral part of our nation's commitment to its future, to its greatness.¹⁹

Johnson himself told a group of astronauts in 1965 that their missions not only increased "our knowledge of technology" but also would lead "to a better life for all." In a 1969 interview, Johnson said that plans to get to the Moon had inspired the country to do something about its educational systems, medical care for the elderly, conservation, and poverty. In his 1971 memoirs, he recalled: "Space was the platform from which the social revolution of the 1960s was launched. We broke out of far more than the atmosphere with our space program. . . . If we could send a man to the moon, we knew we should be able to send a poor boy to school and to provide decent medical care for the aged. In hundreds of other forms the space program had an impact on our lives." A few of the benefits he saw the country reaping from investments in space included pacemakers for heart patients, intercontinental television, lightweight electronics equipment to improve navigation techniques for ships and planes, more abundant food supplies, improved conservation of natural resources, and weather control capabilities that saved lives, crops, and cattle.²⁰

If space exploration tied into Johnson's hopes for a Great Society, it also served his political purposes in the 1964 presidential campaign. Johnson's opponent, the conservative Arizona Senator Barry Goldwater, complained, "We are spending entirely too much money on the manned moon program." He promised that as president he would have "all manned space research . . . directed by the military" and would use the "billions of dollars saved from abandoning the manned lunar program" for "military space missions." As with so many other issues in the campaign, Goldwater was out of sync with the national mood. Polls in the spring and fall of 1964 showed 64–69 percent of the public favorably disposed to landing an American on the Moon, with 78 percent saying the Apollo Program

should be maintained at its current pace or speeded up. Only 20 percent of the country supported space spending strictly for military purposes. In response, Johnson refused "to slacken in our nationally approved effort to reach the moon as soon as we can." Identification with the widely backed Apollo mission was superb politics in an election year.²¹

Johnson and the Budget Crisis

While Johnson gave wholehearted support to Apollo, he also thought about what, if any, big projects might come next. In January 1964, he asked Webb to describe NASA's future plans, specifically asking how "hardware and development programs" would be tied to "prospective missions." In May, Webb provided a tentative answer in which he said that NASA had "virtually completed the investment in facilities" that would land astronauts on the Moon and "meet a broad range of not yet specified tasks." These might include a greater mastery of space science, which would improve weather prediction and control; exploration of the Moon to expand our understanding of the origins of the solar system; a search for life on other planets; the development of space stations, manned and unmanned; better weather, communications, and navigation satellites; and exploration of the near planets and probes of more distant ones.

Not until February 1965, however, did Webb give the president a more precise statement of NASA's future plans. Sensing that Johnson, with expanding commitments at home and abroad, was not eager for new big spending on space, Webb backed away from most of the proposals he had identified in his May 1964 letter. Instead, he urged commitments to two more modest programs: the exploration of Mars through an unmanned landing and further exploration of the Moon with the technology developed for Apollo. The distinguishing features of Webb's proposals, an aide told Johnson, were the absence of a request for any "major new launch vehicle systems" and a continuation of NASA funding at current levels.²²

With Apollo still years away from fulfillment, Johnson was unwilling to make new commitments of any kind. When Webb asked permission to give the chairmen of the House and Senate Space Committees copies of his February letter to inform the Congress about possible future NASA projects, Johnson resisted. "Why do we need to do anything?" he asked in a reply to Jack Valenti, his aide handling the matter. "I would think I would have more leeway & running room by saying nothing[,] which I would prefer."²³

Beginning in 1965, Johnson took a two-track approach to NASA and

space exploration. His only priority was landing a man on the Moon by the end of the decade, as Kennedy and then he had promised. Beyond that, he resisted significant commitments to post-Apollo planning that would cost billions of dollars and engage the country's prestige and energy. One of the striking features of Johnson's memoirs on his presidency is that he devotes only seventeen of six hundred pages to a discussion of space. And of those seventeen pages, only three describe space policy during his presidency. The rest focuses on his Senate and vice presidential years, the period 1957–63, when he felt he had done his most important work for the space program. Indeed, in an interview with Walter Cronkite in 1969, Johnson noted: "Very frankly, I think I spent more time in the space field in '57 and '58 and '59 and '60, and up to '63, than I did after I became President."²⁴

This is not to say that Johnson lost interest in space achievements. He attended closely to the various space missions in 1965–68. As Welsh remembered, Johnson watched each mission on television. "He had the astronauts in to see him at the White House. He had them to the ranch. He followed them with a real sense of personal interest. As a matter of fact, he said that he really in a sense flew with them on every flight from the beginning of the launch till they landed safely." Johnson himself told Cronkite: "I have ridden on every mission. . . . I've watched with eagerness, and pride, their every movement."²⁵

Nevertheless, his interest did not translate into support for post-Apollo projects. Everything that had initially spurred Johnson to back a major American effort in space—fear of Soviet superiority and a desire for economic and political gains—now became reasons to avoid substantial commitments to new big space programs. Johnson's concern, for example, that Soviet advances in space might undermine America's national security and prestige in the Soviet-U.S. competition for global influence steadily faded from view during his presidential years. In the spring of 1966, after the Soviets had landed an unmanned spaceship on the Moon, Webb pressed the president to use the Soviet feat to extract more money for NASA from Congress. Webb told Johnson that he had done his best to "minimize the political risk to your Administration from the fact that we are operating substantially under what would be the most efficient program." This was Webb's way of warning that the Soviets might beat the United States to a manned Moon landing, for which Johnson would pay a high political price.²⁶

But Johnson was not impressed. He was justifiably confident that the United States would land men on the Moon ahead of the Soviets, and he

was certain that Moscow was now more eager for cooperation than competition with the United States in space. Indeed, nine days before Webb's warning about the continuing Soviet threat to America's leadership in space, Johnson had issued "a statement outlining the essential elements of a celestial bodies treaty" and asked U.N. Ambassador Arthur Goldberg to initiate discussions. During the next three months, Soviet and American negotiators drafted nine initial articles of an outer space treaty. By December, additional points of agreement were incorporated into the treaty, which Johnson now publicly described as the "most important arms control development since the Limited Test Ban Treaty of 1963." The space treaty, signed in January 1967 and entered into force in October, banned the placing of weapons of mass destruction in orbit, in outer space, or on celestial bodies, established an unconditional commitment to assist and return astronauts who landed in another country, and forbade claims of sovereignty over celestial bodies.²⁷

To Johnson and the State Department, the agreement meant a "de-fusing of the space race" and a reduction or even an end to "much of the pressure to race for new and distant goals." Henry Owen of the department's Policy Planning Council anticipated "strong opposition" from NASA and the Space Council to additional cooperation with the Soviets in space, because it would mean less funding of post-Apollo projects. More cooperation with Moscow and less ambitious space plans, Owen told Walt W. Rostow, chairman of the Policy Planning Council, "will save money, which can go to (i) foreign aid, (ii) domestic purposes—thus mitigating the strain of the war in Vietnam." Owen urged Rostow to get into the fight with NASA and to enlist "someone on the domestic side of the White House staff . . . to ensure that someone . . . representing the constituency whose interests are most directly affected, gets into the fight." A State Department paper entitled "Space Goals after the Lunar Landing" argued that by deemphasizing or stretching out "additional costly programs aimed at the moon and beyond, resources may to some extent be released for other objectives—foreign aid, domestic needs, scientific efforts in other areas—which might serve more immediate, higher priority U.S. interests."²⁸

Johnson agreed. The increasing costs of the war in Vietnam, which began to escalate rapidly in 1965, and the outlays for the antipoverty and Great Society programs, which also made substantial budgetary demands beginning in 1965, were central considerations in Johnson's resistance to post-Apollo space commitments. In July 1969, at the end of his administration and after the successful Moon landing, Johnson was vague and evasive about post-Apollo plans. "What would you like to see as the next

space goal?" an interviewer asked him. "I don't want to be setting goals for those that are responsible for this effort," he said. "I would like to take all that we have done and be sure that we utilize all the knowledge that we have gained up to now, and to follow through to milk the entire Apollo program of every benefit that can come from it." Johnson then ticked off the various ideas others had for post-Apollo planning: space stations, additional Moon shots, studies in space medicine, and unmanned trips to other planets. Personally, he would not say what he favored but hoped that we would continue to have a vigorous space program.²⁹

Johnson's remarks were symptomatic of his refusal to make significant, large-scale commitments beyond Apollo in his 1967-69 budgets. His rhetoric masked the battles he and Webb had fought over funding for NASA's future. After suffering a modest cut of about \$75 million in 1966, Webb was determined to increase NASA's funding in 1967. But Johnson wouldn't hear of it. Webb's request for \$5.3 billion could not withstand a \$300 million reduction. In accepting the president's cut, Webb warned against keeping NASA's funding at the current level for another year. "The 1968 budget will be a major turning point with indicated requirements on the order of \$6 billion of new obligational authority," he told Johnson in May 1966. By August, however, it was clear that Congress and the president would drop NASA's funding below \$5 billion for FY 1967. This would "leave no choice," Webb berated Johnson, "but to accelerate the rate at which we are carrying on the liquidation of some of the capabilities which we have built up." He predicted that options would now be foreclosed and that doubt and uncertainty would demoralize NASA. And, he bluntly declared, "There has not been a single important new space project started since you became President. Under the 1968 guidelines very little looking to the future can be done next year. . . . I cannot avoid the feeling that this is not in the best interests of the country."³⁰

Johnson relied on his budget director, Charles Schultze, to counter Webb's assertions. Schultze argued that NASA's funding was entirely adequate to meet the 1969 deadline for a Moon landing and to work toward more distant goals, such as a Mars landing and/or Earth orbital stations. After all, "the space program is not a WPA," Schultze declared. Neither he nor Johnson felt that NASA's budget was skimpy alongside the \$2 billion in spending on elementary and secondary education, \$1.8 billion on the poverty program, \$200 million on water pollution control, and \$25 million for high-speed ground transportation. A \$5 billion space budget or even a little below that would not "wreck the space program," Schultze contended, nor would it lead to "the liquidation of some of the capabili-

ties we have built up." NASA's funding did not represent "a lack of support for the space program." Moreover, Schultze did not see how in the context of the fighting in Vietnam the administration could afford to meet Webb's request. Johnson agreed with Schultze and convinced Webb to back his decision publicly, though privately the NASA administrator continued to press his case, unsuccessfully requesting an additional \$182 million above the \$455 million slated for post-Apollo planning.³¹

Johnson saw little political risk in turning aside Webb's demands for more money. By the end of 1966, it was clear to him that NASA and space exploration beyond the Apollo landing had diminished popular appeal. By the summer of 1965, a third of the nation favored cutting the space budget, while only 16 percent wanted to increase it. Over the next three and a half years, support for cutting space spending went up to 40 percent, with those preferring an increase dropping to 14 percent. A poll taken in the summer of 1969 found that 53 percent of the country was opposed to a manned mission to Mars. At the end of 1967, the *New York Times* reported that a poll conducted in six American cities showed that five other public issues held priority over efforts in outer space. Residents of these cities preferred tackling air and water pollution, job training for unskilled workers, national beautification, and poverty before spending additional federal funds on space. The following year *Newsweek* echoed the *Times*'s findings, stating, "The U.S. space program is in decline. The Viet Nam war and the desperate conditions of the nation's poor and its cities—which make space flight seem, in comparison, like an embarrassing national self-indulgence—have combined to drag down a program where the sky was no longer the limit."³²

In addition, Congress was strongly disposed to reduce NASA's budget. A White House survey of congressional leaders at the end of 1966 revealed pronounced sentiment for keeping Apollo on track but for cutting NASA spending by skimping on post-Apollo outlays. In this context, Johnson's request in January 1967 for a \$5 billion NASA budget for FY 1968, including \$455 million for post-Apollo programs, was pretty bold.³³

Yet Johnson's inclination to be generous with NASA and provide for a modest amount of post-Apollo spending could withstand neither a disastrous fire in an Apollo command module in January 1967 nor a growing U.S. budget deficit spurred by the fighting in Vietnam. On January 27, a fire destroyed the *Apollo 1* capsule and killed astronauts Roger B. Chafee, Edward H. White III, and Virgil I. Grissom during a test at Cape Kennedy. In addition to the tragic loss of life, the fire undermined national confidence in NASA, which was now accused of carelessness in trying to move Project

Apollo forward too quickly. The fire, Johnson said later, represented "an all-time low" for the space effort. "I grieved [not only] for the men and their families but [also] . . . for the space organization. I felt very sad and sorry for Jim Webb and all of his loyal employees." Senate hearings raised questions about a great many defects in the spacecraft and brought Webb into sharp conflict with three senators, who saw him as whitewashing NASA's failings. The *New York Times*, which was also highly critical of Webb, said that NASA stood for "Never a Straight Answer." Though the hearings were "unpleasant and embarrassing for NASA . . . on the whole," an administrative history of the agency asserts, "they gave NASA a sympathetic forum in which to explain how a tragedy had come about, and show how it would serve to correct deficiencies." NASA's forthrightness in responding to the failings that produced the fire restored a measure of confidence in the agency and prompted the Senate committee to recommend that NASA continue to move the Apollo Program forward to achieve its goal.³⁴

A federal budget crisis in the summer of 1967 dealt NASA another blow. A \$29 billion deficit brought on by Vietnam spending persuaded Johnson to ask Congress for a 10 percent increase in income taxes. To persuade Congress, Johnson felt compelled to match the tax increase with spending cuts applied to FY 1968 beginning in October 1967. NASA was targeted for \$500 million in reductions. Webb objected that, with NASA "just now getting back up to speed after the interruptions and difficulties associated with the accident," it would be "the straw that breaks the camel's back," meaning, "the momentum we have achieved will be lost." For Johnson, there was no choice, except where to apply the cuts in NASA programs. As before, despite recommendations to the contrary, he stuck to keeping Apollo on schedule, agreeing instead to center cuts on post-Apollo applications and the unmanned landing on Mars. Once again, Webb and NASA had to accommodate themselves to a reduced budget, now, \$4.59 billion. In spite of everything, Webb was still able to assure Johnson that "the goal of the manned lunar landing in this decade is preserved."³⁵

The cuts genuinely troubled Johnson. Whenever there were reductions, he would tell Webb, "Next year I hope to make up for this." Johnson "had almost supreme confidence that at some point he could give us resources again and that we could catch up," Webb recalled. More specifically, in a message to Webb in September 1967, the president asked him to "be sure to make abundantly clear [to a congressional committee] that I do not choose to take one dime from my budget for space appropriations for this year." The "Congress forced me to agree to effect some reductions or lose

the tax bill." While Johnson's message was partly a case of political finger-pointing, he was truly uncomfortable reining in NASA or any government program he believed served the national well-being. He loved to quote Speaker Sam Rayburn's adage: "Any jackass can kick a barn down, but it takes a good carpenter to build one." More to the point, his whole political career had been focused on building and using government programs to expand the economy, raise living standards, relieve privation, and build his Great Society. Overreaching himself by trying to institute domestic reforms and fight a war at the same time, he could not find the means to spend simultaneously on guns and butter. It was a reality he found difficult to accept.³⁶

Webb also struggled against the reality of declining commitments to NASA. In November 1967, he pressed Johnson's budget director to urge a strong statement by the president about NASA funding when signing its appropriation bill. NASA's congressional backers, Webb said, saw the president as having "knifed" the very activities he had previously been urging them to support." Johnson's inconsistent leadership, Webb added, had created "bitter feelings . . . in a number of quarters." Webb wanted Johnson to emphasize the continuing Soviet danger to American preeminence in space and to say: "Although we will at this time have to postpone important parts of our space program, let this fact be clear: We are fixed in our resolve to master the challenge of space."³⁷

Johnson and Webb

Yet nothing Johnson said could change the reality of shrinking budgets and flagging enthusiasm for space exploration after Apollo. A *New York Times*'s story in April 1968 observed:

After a heady decade of uninterrupted hiring, building and dreaming great dreams of far-reaching exploration, the American space program is gearing down to a slower pace and a less certain future. . . . The growing feeling in the space establishment is that once astronauts have landed on the moon, they will have no other place of significance to go for several years because of sharp budget cuts. These cuts have trimmed to the bone all preparations for future missions. It is as if the astronauts are heading for a dead-end on the moon.

By September, after the White House had proposed to reduce NASA spending another quarter of a billion dollars and congressional appropriations committees penciled in only \$3.99 billion for NASA in FY 1969, Webb resigned.³⁸

There are conflicting accounts of the reasons behind Webb's departure.

Webb himself claimed that he wanted to get out before the *Apollo 7* and 8 missions, the manned orbits of Earth and the Moon in the fall of 1968, so that he could respond to any failure by going after critics in and out of the Congress. Others indicated that Webb was surprised when Johnson accepted his resignation. According to these accounts, Webb had used the threat of resignation repeatedly with the president as a way to press NASA's case. On September 16, 1968, however, Johnson, who had gotten "fed up with this same old story," took Webb up on the offer, saying, "Let's call in the press."³⁹

Whatever the realities behind the decision, Webb tried to turn his resignation to NASA's advantage. At a press conference on September 16, he "bitterly" complained that "congressional budget cuts had put the United States second in the space race" behind the Soviets. Though he denied that he was leaving because of reductions totaling \$1.4 billion over the previous four years, he nevertheless said that "the agency had been used 'as a sort of whipping boy' by Congress and other agencies competing for Federal funds. And he made it clear that he felt the U.S. is still behind the Soviet Union in space, and that the reason for this second place is a lack of funds." Sources told the *Washington Post* that "Webb was discouraged by the budget cuts, tired of fighting to have Congress restore them and even wearier of debating the urgency of the space program with the Bureau of the Budget and President Johnson."⁴⁰

Webb's public comments provoked an angry response from the administration. Donald F. Hornig, the president's science advisor, sent Johnson a memo describing Webb's assertions as "unconscionable statements" that "were undoubtedly motivated" by NASA's "budgetary problems." Hornig disputed assertions that the Soviets had "a capability that could change the basic structure and balance of power in the world," that the U.S. was clearly second in space and that a Soviet manned lunar landing could be achieved in the next year—a time scale that is competitive with, or ahead of Apollo." Hornig thought the United States was at least a year ahead of the Soviets and, if the president agreed, he wished to press Webb and others at NASA to set the record straight. Welsh at the Space Council also felt that Webb's estimate of Soviet space capabilities was "inaccurate" and that the U.S. missions in space had demonstrated American preeminence.⁴¹

But Johnson sided with Webb, inviting him to respond to Welsh and telling his aides to instruct Hornig not to get into the NASA debate. In a response to Hornig's memo, Johnson said: "Drop it! That is my feeling, but get Jim Webb to get me a prompt reply—all his scientists—all his private ones—to support him and me." In a written memo to Hornig, which

Johnson wanted conveyed only on the phone, the president said, "It is hard for me to believe that Jim Webb would make 'unconscionable statements' or be 'motivated' entirely by budgetary problems." Johnson defended Webb's concern that "the trend of the Soviet program upward and the U.S. program downward" might allow the Soviets "to achieve both the image and reality of power and forward motion." Johnson warned Hornig that "even if your group should develop evidence to sustain their views, your report might be shortly followed by some tragic occurrence in the U.S. program or a major triumph in the Russian one. This would inevitably bring into question the judgment of your group in a way that might impair its usefulness." Johnson also identified himself with Webb's complaints about budget shortfalls, saying, "It was only with great reluctance that for the past two years I have taken action to meet the overall fiscal requirements laid down by a determined group in the Congress by accepting cuts made in the House Appropriations Committee."⁴²

Nonetheless, Webb's concerns were greatly exaggerated, as demonstrated by the successful *Apollo 7* and 8 missions in September and December, respectively. But Johnson backed him nevertheless. Partly, he had a warm feeling for Webb, who had served him so loyally for almost five years. To Johnson, this was no small consideration. In February 1968, with the Tet offensive in Vietnam in full swing and the White House besieged by antiwar protests, Johnson talked to Thomas O. Paine about becoming deputy administrator of NASA. As Paine recalled, Johnson stressed the problems faced by his administration, "how much he needed people to come into the government and shoulder part of this burden and relieve him of it, to serve him loyally, help him move the country ahead in these very difficult times." In addition, Johnson liked and admired Webb. After a conversation with Webb about the *Apollo* fire, Johnson remembered telling Mrs. Johnson: "I know now why Jim Webb was an old Marine and a good one. He's got the courage. He goes through a disaster like this and he says, 'We just got to go on and do what we know is right.' . . . And he did."⁴³

But more than personal sentiment determined Johnson's support of Webb. He felt that the historical reputation of his administration was in part at stake. If Webb was right about the potential for renewed Soviet dominance in space, if Moscow beat the United States in the Moon race, Johnson believed that he and Webb would be seen as having presided over a failed or at least inadequate space program. By letting Webb beat up on Congress for shortchanging NASA, Johnson was preparing to point the finger at representatives and senators for any retrospective weaknesses

historians saw in the Johnson administration's space effort. Shortly after the *Apollo 7* and 8 successes, when Johnson gave Webb NASA's Distinguished Service Medal and praised him as "the best administrator in the Federal Government," he was leaving no doubt that, unlike many in the Congress, he had been an ardent advocate of NASA in general and of Apollo in particular. Johnson wanted to be remembered as a president who made his mark in space.⁴⁴

Conclusion

Johnson's historical reputation as a senator, vice president, and president will never be more than marginally affected by his part in the development of America's space program. His initiatives as an "imperial president" will always identify him more with domestic reforms, such as civil rights, Medicare, federal aid to education, and other war on poverty and Great Society measures, than with Project Apollo. These initiatives will stand with the disaster in Vietnam as the centerpieces of his political career much more than his presidential goals in space. Nevertheless, in time, as the United States progresses into the space age and ever more important discoveries emerge about our universe, Johnson will stand in the front rank of those who had the foresight and determination, as well as the skill to use presidential power—in spite of its undeniable limitations—to initiate America's probe into the farthest reaches of outer space. For this driven, almost madly ambitious man from rural south central Texas, it may be fairly said that in his lifetime he both figuratively and literally reached for the Moon.

Notes

1. This introductory material is from Robert Dallek, *Lone Star Rising: Lyndon Johnson and His Times* (New York: Oxford University Press, 1991), pp. 529–34.
2. See Rowland Evans and Robert Novak, *Lyndon B. Johnson: The Exercise of Power* (New York: Random House, 1966), p. 323; Doris Kearns, *Lyndon Johnson and the American Dream* (New York: Harper and Row, 1976), pp. 162–64; Arthur M. Schlesinger Jr., *Robert Kennedy and His Times* (Boston: Little, Brown, 1978), pp. 622–23.
3. This material is drawn from Lyndon B. Johnson, *The Vantage Point: Perspectives of the Presidency, 1963–1969* (New York: Random House, 1971), pp. 278–79; and Dallek, "The Most Insignificant Office," chap. 1 in *Splendid Misery: Lyndon Johnson and His Times, 1961–1973* (forthcoming from Oxford University Press).

4. Kennedy to Johnson, Apr. 20, 1961; Johnson to Kennedy, Apr. 28, 1961, President's Office Files (POF), John F. Kennedy Library (JFKL), Boston, Mass.; George Reedy, memorandum, n.d. [clearly 1961], beginning: "Despite all the heated public discussion," Office Files, George E. Reedy (OF/GER), Lyndon B. Johnson Library (LBJL), Austin, Texas; Walter A. McDougall, . . . *The Heavens and the Earth: A Political History of the Space Age* (New York: Basic Books, 1985), p. 320; Edward C. Welsh, interview, July 18, 1969, JFKL.
5. McDougall, . . . *The Heavens and the Earth*, pp. 302–3.
6. Ibid., pp. 308–10; Newton Minow, interview, Mar. 19, 1971, LBJL.
7. Kennedy to Johnson, July 29, 1963; Johnson to Kennedy, July 31, 1963, White House Famous Names (WHFN), LBJL; Minow, interview; McDougall, . . . *The Heavens and the Earth*, pp. 322–23, 376, 389–96.
8. Johnson to Welsh, July 26 and 29, 1961; Johnson to Walter Jenkins, Aug. 2, 1961; Welsh to Johnson, Aug. 11, Sept. 8, 13, and 14, 1961; Welsh to Chelsea L. Henson, Oct. 9, 1961, Vice Presidential Papers (VPP), LBJL.
9. George Smathers, interview, Aug. 1–Oct. 24, 1989, Senate Historical Office (SHO), Washington, D.C.; McDougall, . . . *The Heavens and the Earth*, pp. 361–63, 373–76.
10. Richard L. Callaghan, phone conversation with Robert Sherrod, Jan. 28, 1971; Kenneth O'Donnell, phone conversation with Sherrod, May 13, 1971, Robert Sherrod Apollo Collection (RSAC), NASA Historical Reference Collection, NASA History Office, Washington, D.C. Johnson is quoted by Leo Janos in "The Last Days of the President: LBJ in Retirement," *Atlantic*, July 1973, pp. 35–41.
11. Webb, interview by James Burke, May 23, 1979, James Webb Papers, NASA Historical Reference Collection.
12. "Questions from Everett Collier re President Johnson's Attitude on Space," Feb. 8, 1964, White House Central File (WHCF), EX105; "Administrative History of NASA" (unpublished manuscript, 1966), pp. 1–9, LBJL; Robert A. Divine, "Lyndon B. Johnson and the Politics of Space," in *The Johnson Years*, vol. 2: *Vietnam, the Environment, and Science*, ed. Robert A. Divine (Lawrence: University Press of Kansas, 1987), pp. 233–34.
13. The Johnson quote is in Divine, *Johnson Years*, p. 234; Kennedy's view is in "Administrative History of NASA," p. 3.
14. Stuart H. Loory, "U.S. Talking Down the Space Race," *New York Herald Tribune*, June 9, 1964, clipping in "NASA Current News folder," NASA Historical Reference Collection; Charles Johnson to McGeorge Bundy, Feb. 4, 1964; Bundy to Johnson, Feb. 29, 1964; Webb to Johnson, Apr. 30 and June 29, 1964; National Security Action Memoranda (NSAM) 285 and 288, National Security Files, LBJL; John Glenn, interview, Oct. 18, 1968, RSAC, NASA Historical Reference Collection.
15. See Merle Miller, *Lyndon* (New York: Simon and Shuster, 1980), p. 410; Kearns, *Lyndon Johnson*, p. 412n3; Vaughn Davis Bornet, *The Presidency of Lyndon B. Johnson* (Lawrence: University Press of Kansas, 1983), p. 8.

16. Webb to Johnson, Dec. 20, 1963, WHCF, GEN, FG260, NASA; Webb to Johnson, Oct. 29, 1965, WHCF, EX, FG260, both in LBJL.
17. See the weekly NASA reports to Johnson from Welsh for 1964 in WHCF, EX\OS, LBJL; and the Webb-Johnson briefing, Oct. 14, 1968, from which the quotations are drawn, in James Webb Papers, NASA Historical Reference Collection.
18. Johnson, "Remarks at the University of Michigan," May 22, 1964, *Public Papers of the Presidents of the United States: Lyndon B. Johnson, 1963-64*, 10 vols. (Washington, D.C.: Government Printing Office, 1965), 1:704-7.
19. Horace Busby to Welsh, May 13, 1964, White House Aides, Horace Busby Papers, LBJL; Webb to Johnson, Nov. 30, 1964; "Introductory Comments," Nov. 12, 1965, quoting Humphrey's speech of Mar. 19, 1965, LBJ Files, NASA Historical Reference Collection.
20. Divine, *Johnson Years*, p. 235; Johnson, interview by Walter Cronkite, July 5, 1969, LBJ Files, NASA Historical Reference Collection; Johnson, *Vantage Point*, pp. 285-86.
21. Welsh to Johnson, May 2, July 21, Sept. 2, and Oct. 24, 1964, WHCF, EX\OS, LBJL; Divine, *Johnson Years*, pp. 234-35.
22. Johnson to Webb, Jan. 30, 1964; Webb to Johnson, Feb. 16, 1965; Jack Valenti to Johnson, Feb. 17, 1965, WHCF, EX\OS; Webb to Johnson, May 20, 1964; Charles S. Sheldon II to George Reedy, May 26, 1964, White House Aides, Bill Moyers Papers, LBJL.
23. Webb to Valenti, Mar. 30, 1965; Valenti to Johnson, Mar. 30 and Apr. 1, 1965; Johnson's note is handwritten on Valenti's March 30 memo, WHCF, EX\OS, LBJL.
24. Johnson, *Vantage Point*, pp. 270-86; Johnson, interview.
25. Welsh, interview; Johnson, interview.
26. Webb to Johnson, Apr. 1 and May 16, 1966, WHCF, Confidential File (CF), OS, LBJL.
27. See "Outer Space Treaty Chronology" in White House Aides, Joseph Califano Papers; and the materials for 1966-67 in WHCF, Legislative Background, Outer Space History, LBJL.
28. Owen to Rostow, Dec. 9, 1966; "Space Goals after Lunar Landing," Oct. 1966, NSF, Subject Files, LBJL.
29. Johnson, interview.
30. See Divine, *Johnson Years*, pp. 238-40; Webb to Johnson, May 16, 1966, WHCF, CF, OS; Webb to Johnson, Aug. 26, 1966, WHCF, EX\OS, LBJL.
31. Schultze to Johnson, Sept. 1 and 20, 1966, WHCF, EX\OS; Webb to Johnson, Dec. 17, 1966, WHCF, EX, FI4\FG260, LBJL; Divine, *Johnson Years*, pp. 240-42.
32. Gallup Poll, *Public Opinion, 1935-1971*, vol. 3, 1959-1971 (New York: Random House, 1972), pp. 1952, 2183-84, 2209; "Poll Finds a Drop in Space Interest," *New York Times*, Dec. 3, 1967, clipping in "NASA Current News folder," Dec. 1967, NASA Historical Reference Collection; *Newsweek* is quoted in "Administrative History of NASA," p. 48.

33. For congressional opinion, see the survey made by Charles Schultze in box 23, WHCF, EX\FI4; Johnson's budget is described in "Administrative History of NASA," pp. 17-19.
34. "Administrative History of NASA," pp. 47-52; Johnson, interview; Senator Clinton P. Anderson, interview by Sherrod, July 25, 1968; Sherrod to John B. Oakes, May 24, 1972, RSAC; Welsh, interview by Eugene M. Emme, Feb. 20, 1969, all in NASA Historical Reference Collection.
35. Webb to Johnson, July 10, 1967, WHCF, CF, Agency Reports, NASA; Webb to Johnson, Aug. 10, 1967, WHCF, EX\FI4\FG200, LBJL; Divine, *Johnson Years*, pp. 243-45.
36. Webb, interview, Apr. 29, 1969, LBJL; Johnson's message to Webb is in Divine, *Johnson Years*, pp. 244-45; the Rayburn quotation is in Dallek, *Lone Star Rising*, p. 427.
37. Webb to Schultze, Nov. 6, 1967, WHCF, EX\FI4, FG260, LBJL.
38. John Noble Wilford, "U.S. Space Effort Is Shrinking after Era of Growth," *New York Times*, Apr. 16, 1968, clipping in "NASA Current News folder," Apr. 1968, NASA Historical Reference Collection; Divine, *Johnson Years*, pp. 238, 245.
39. Webb, interview, Apr. 29, 1969; Thomas O. Paine, interview, Mar. 25, 1969, LBJL; Robert Sherrod, memo on Thomas O. Paine, Aug. 25, 1970; David Williamson Jr., interview by Sherrod, Apr. 10, 1972, James Webb Folder, RSAC, NASA Historical Reference Collection.
40. "Administrative History of NASA," pp. 34-37, 53-55; Carroll Kilpatrick and Thomas O'Toole, "Webb Retires Oct. 7 as U.S. Space Chief," *Washington Post*, Sept. 17, 1968, clipping in "NASA Current News folder," Sept. 1968, NASA Historical Reference Collection.
41. Hornig to Johnson, Sept. 26, 1968; Welsh to Johnson, Sept. 30, 1968, WHCF, EX\OS, LBJL.
42. Webb to Johnson, Oct. 1, 2, and 5, 1968, answering Welsh; Johnson, memorandum, n.d.; Johnson to Hornig, memorandum, n.d., both attached to Larry E. Temple, memorandum for the files, Oct. 10, 1968, WHCF, EX\OS, LBJL.
43. Paine, interview; Johnson, interview.
44. Johnson's awards to and praise for Webb are described in Divine, *Johnson Years*, pp. 246-47.